NETWORKWALKS B082 — WEEK 2 PROJECT REPORT

W2-PM5: Zenmap Network Scanning

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Project	Week 2 — PM5	**Date**	11 November 2026
Operating System	Kali Linux	**Host Computer**	MacBook M1
Scanning Tool	Zenmap / Nmap 7.99		

1. Introduction

This practical exercise focused on network discovery and host identification using Zenmap, the graphical interface for Nmap.

The objective was to identify the local network configuration, discover active hosts, record their IP and MAC addresses, and generate a network topology diagram.

The exercise was conducted in my authorised Kali Linux virtual lab environment.

2. Objectives

The objectives were to:

1. Identify the local IP address and network subnet.
2. Identify the default network gateway.
3. Test connectivity to the gateway.
4. Discover active hosts on the local subnet.
5. Count the live hosts discovered.
6. Record the IP and MAC addresses of identified hosts.
7. Perform network discovery using Zenmap.
8. Save the scan results and export the network topology as a PDF.

3. Tools and Environment

The following tools and environment were used:

- Kali Linux running in a virtual machine.
- Zenmap graphical network scanning interface.
- Nmap 7.99.
- Linux terminal.
- ip networking commands.
- ping connectivity utility.

Note on the environment: The project guide referenced a Windows PC. However, I used Kali Linux on my MacBook M1 because I did not have access to a Windows computer. Zenmap was available and successfully installed in my Kali environment.

4. Methodology and Practical Findings

Task 1: Identify the Local IP Address

Command: `ip -br addr`

This command displays network interfaces and their assigned IP addresses in a brief format.

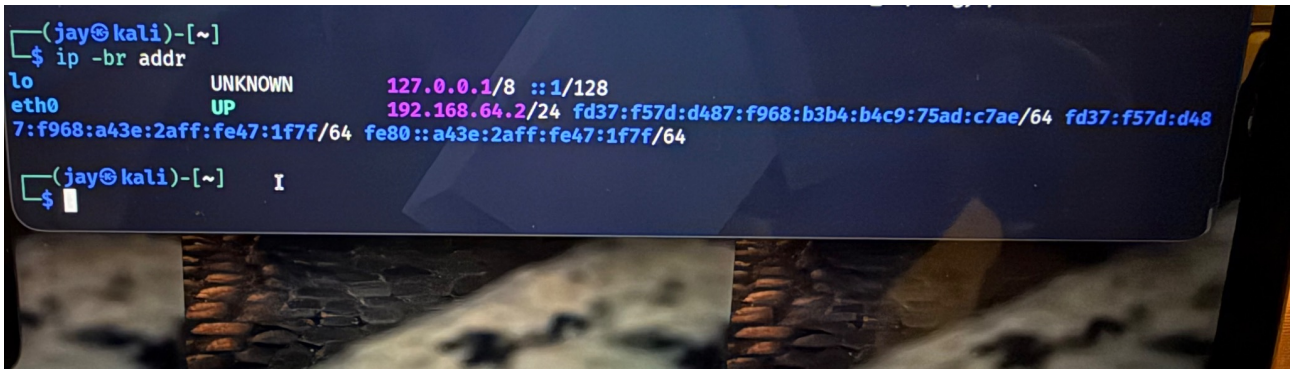
Observed result:

- Network interface: eth0
- Interface status: UP
- IPv4 address: 192.168.64.2/24

The /24 prefix indicates the subnet mask 255.255.255.0.

The identified network was: **192.168.64.0/24**.

Task 1 evidence — local IP address and interface status



Screenshot showing the output of `ip -br addr`, including eth0 in the UP state and IPv4 address 192.168.64.2/24.

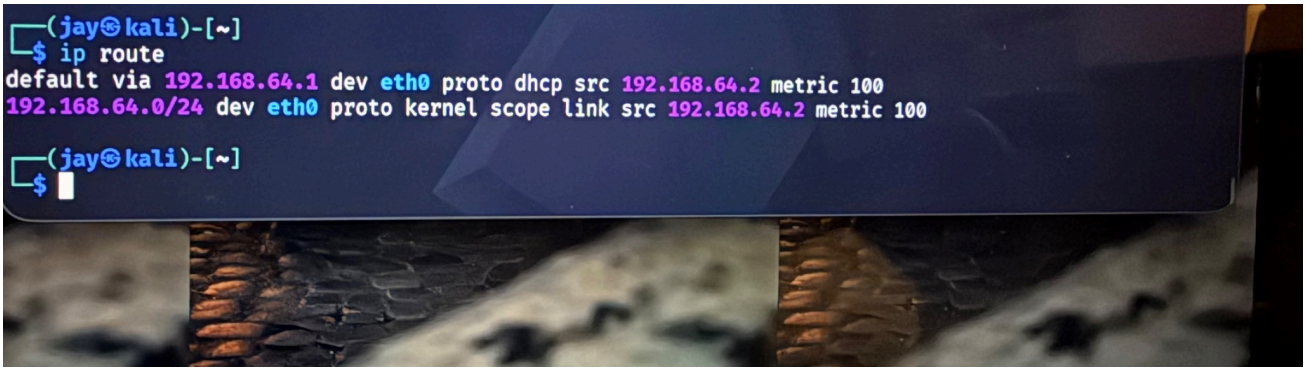
Task 2: Identify the Default Gateway

Command: `ip route`

Observed result:

- Default gateway: 192.168.64.1
- Local network: 192.168.64.0/24
- Interface: eth0

The default gateway provides the route through which the Kali virtual machine communicates with networks outside its local subnet.



Task 2 evidence ? output of the ip route command showing the default gateway 192.168.64.1 and local subnet 192.168.64.0/24.

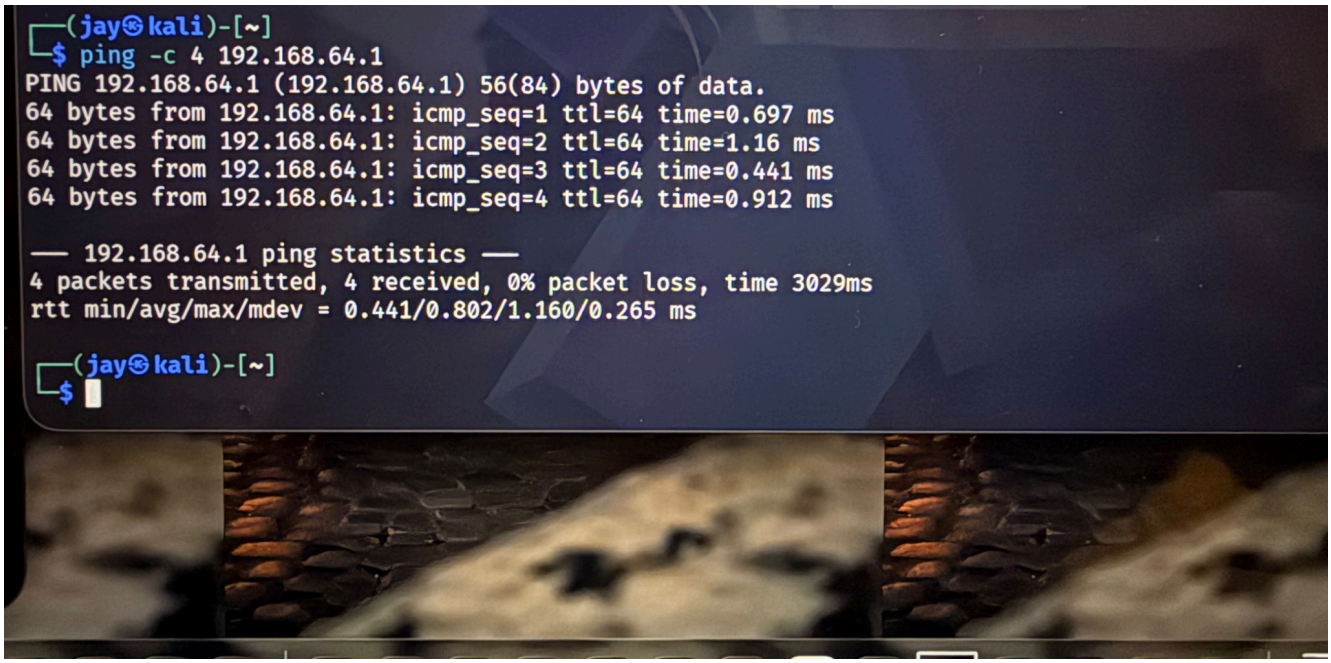
Task 3: Test Gateway Connectivity

Command: `ping -c 4 192.168.64.1`

Observed result:

- Packets transmitted: 4
- Packets received: 4
- Packet loss: 0%
- Average response time: 0.802 ms

Analysis: The gateway responded successfully to all four ICMP echo requests. This confirmed connectivity between the Kali virtual machine and its gateway.



Task 3 evidence ? successful ping test to 192.168.64.1, with four packets received and 0% packet loss.

Task 4: Discover Live Hosts

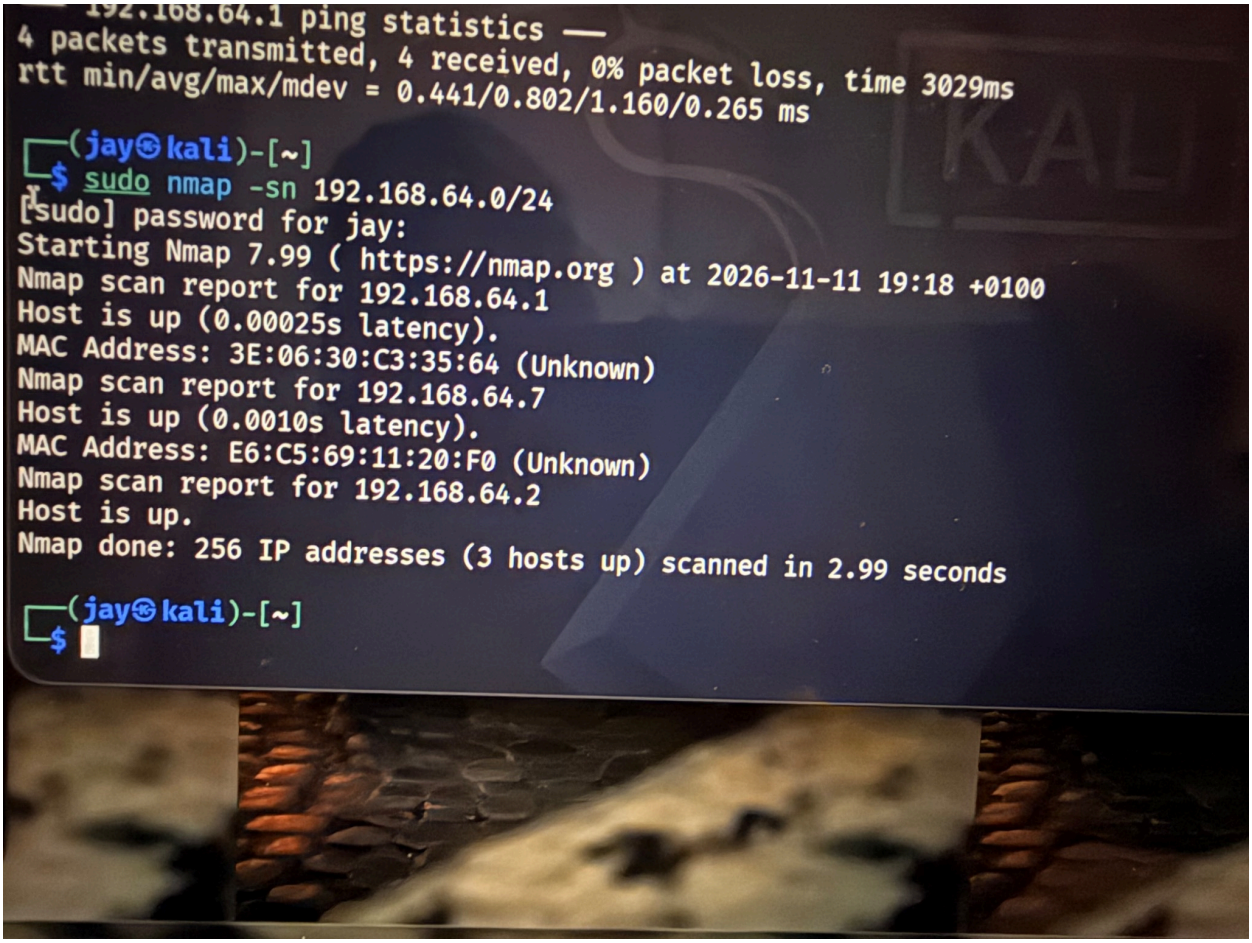
Command: sudo nmap -sn 192.168.64.0/24
The -sn option performs host discovery without conducting a port scan.

Observed result: Nmap scanned 256 IP addresses and identified three live hosts.

IP address	Status	MAC address
192.168.64.1	Up	3E:06:30:C3:35:64
192.168.64.7	Up	E6:C5:69:11:20:F0
192.168.64.2	Up	Identified separately below

Total live hosts discovered: 3

The address 192.168.64.1 was identified as the gateway, while 192.168.64.2 was the Kali virtual machine. The identity of 192.168.64.7 was not independently established.



Task 4 evidence ? Nmap host discovery scan showing 256 IP addresses scanned and three live hosts identified.

Task 5: Identify MAC Addresses

Command: `ip neigh`

This command displays entries in the neighbour table, including IP-to-MAC address mappings.

The output confirmed the MAC addresses of:

- 192.168.64.1 — 3E:06:30:C3:35:64
- 192.168.64.7 — E6:C5:69:11:20:F0

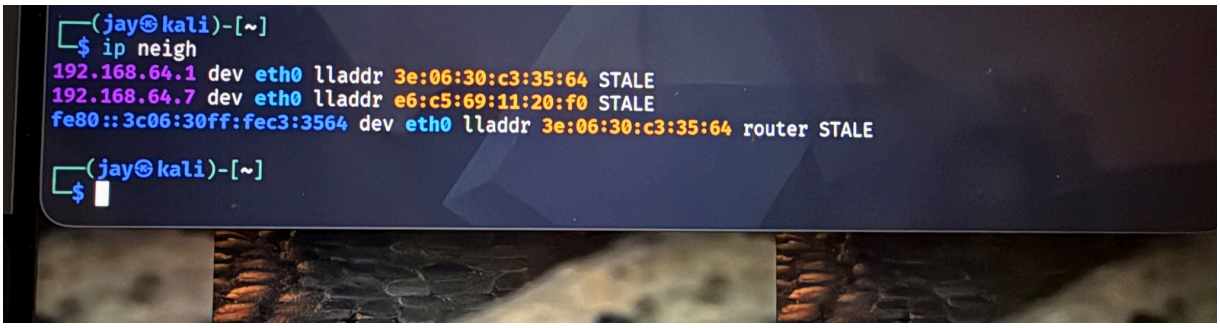
The Kali virtual machine's MAC address was obtained separately using:

`ip link show eth0`

Observed Kali MAC address: A6:3E:2A:47:1F:7F

The completed device inventory is:

Device	IP address	MAC address
Gateway	192.168.64.1	3E:06:30:C3:35:64
Other live host	192.168.64.7	E6:C5:69:11:20:F0
Kali virtual machine	192.168.64.2	A6:3E:2A:47:1F:7F



Task 5 evidence · ip neigh output showing the IP-to-MAC address mappings for the gateway and other live host.

Task 6: Perform Network Scanning with Zenmap

Zenmap was launched using:

`sudo zenmap`

The following configuration was used:

· Target: 192.168.64.0/24

· Profile: Ping Scan

The scan successfully identified the same three live hosts previously discovered using the Nmap terminal command.

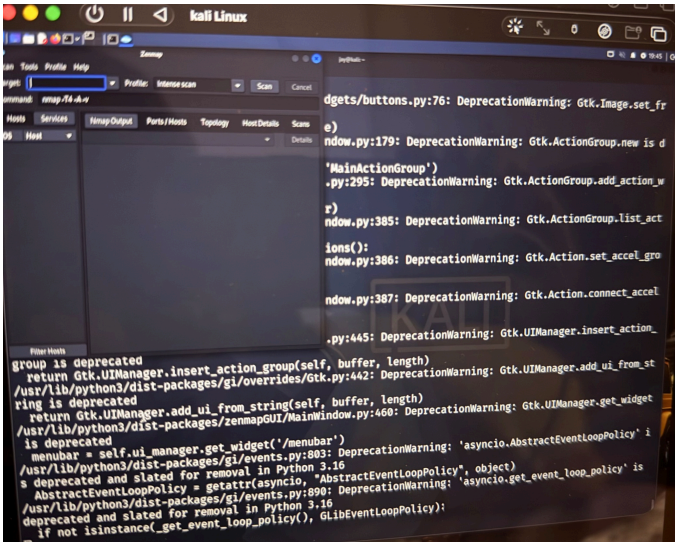
Result: Three hosts were reported as up.

The scan results were saved in XML format as:

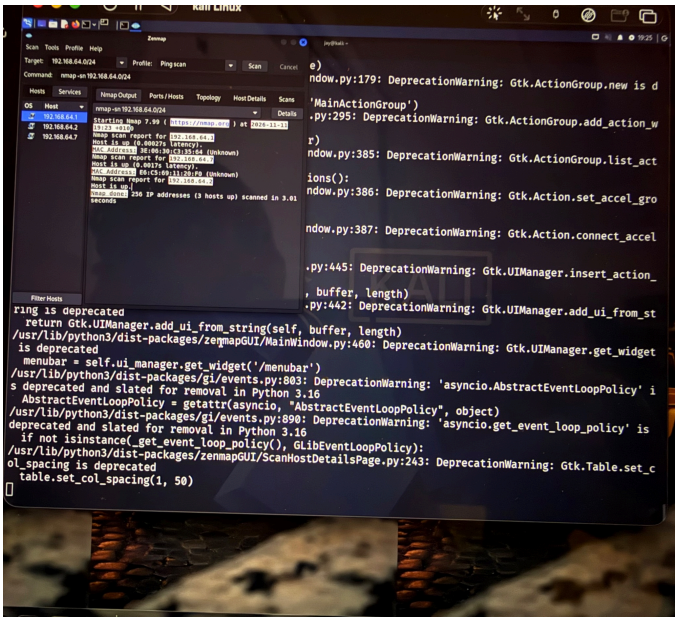
`PM5_Zenmap_Scan_Results.xml`

Task 6 evidence · Zenmap Network Scanning

The screenshots below document the Zenmap scan setup and its results.



Evidence 1 ? Zenmap opened with the scan profile interface displayed.

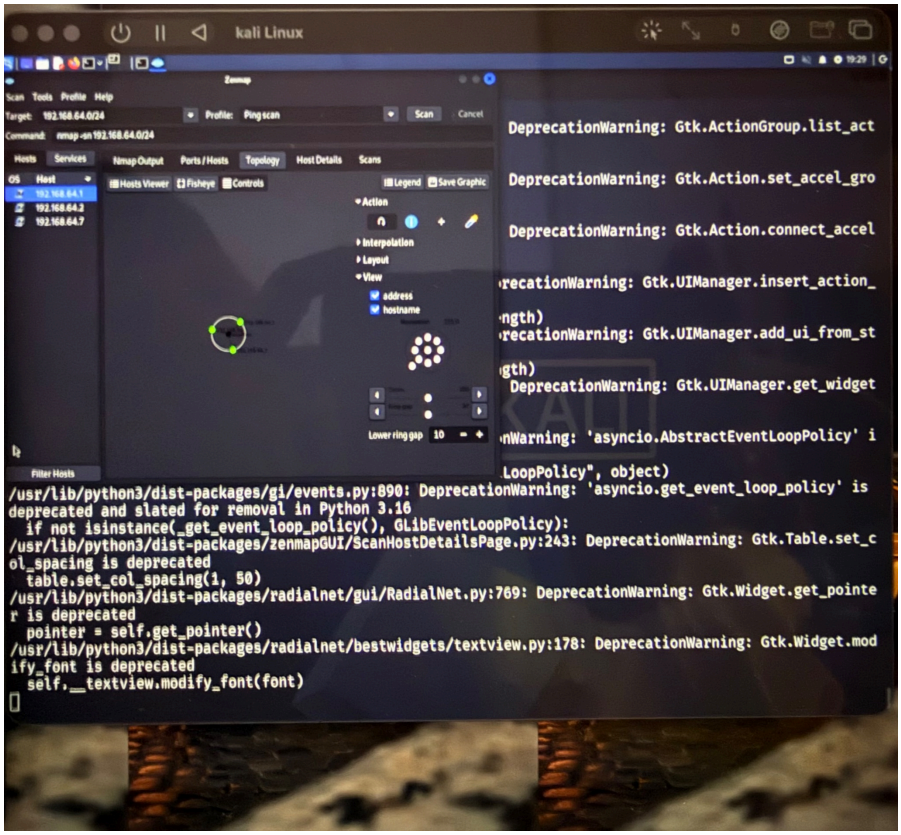


Evidence 2 ? Ping Scan results for 192.168.64.0/24, showing three live hosts.

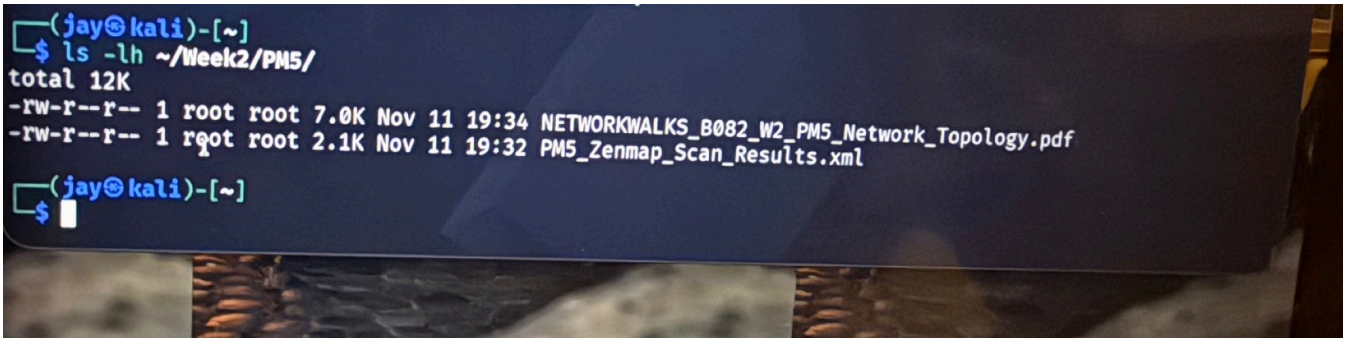
Task 7: Generate Network Topology

The Topology tab in Zenmap was used to view the network discovery results. The network topology was exported as a PDF file named:

NETWORKWALKS_B082_W2_PM5_Network_Topology.pdf



Evidence 1 ? Zenmap Topology tab displaying the discovered network layout.



Evidence 2 ? Terminal listing confirming that the network topology PDF and Zenmap XML scan results were saved in the PM5 folder.

The exported PDF was opened and checked, and its text was readable.

Activity	Result
Local IPv4 address	192.168.64.2
Network subnet	192.168.64.0/24
Default gateway	192.168.64.1
Gateway connectivity	Successful
Packet loss	0%
IP addresses scanned	256
Live hosts discovered	3
Zenmap scan	Successful
XML results	Saved
Network topology PDF	Exported and verified

6. Challenges Encountered

1. The assignment guide referenced Windows, but I used a MacBook M1 with Kali Linux because a Windows computer was unavailable.

2. The Zenmap topology display initially had black text on a black background, making it difficult to read.

3. The Kali machine's MAC address was not displayed in the Nmap host discovery output, so I used ip link show eth0 to identify it separately.

These challenges were addressed sufficiently to complete the practical activities and save the required scan and topology files.

7. Learning Outcomes

Through this exercise, I gained practical experience in:

- Identifying network interfaces and IP addresses.
- Understanding subnet prefixes and default gateways.
- Testing network connectivity using ping.
- Discovering active hosts using Nmap.
- Understanding the difference between IP addresses and MAC addresses.
- Using Zenmap's graphical interface.
- Saving network scan results in XML format.
- Generating and exporting a network topology PDF.

8. Conclusion

The W2-PM5 network scanning practical was completed successfully using Kali Linux and Zenmap.

The exercise identified the local network configuration, confirmed gateway connectivity, and discovered three live hosts on the authorised virtual network.

The IP and MAC address information was documented, the Zenmap scan results were saved in XML format, and the network topology was exported to a readable PDF.

This practical strengthened my understanding of network discovery, host identification and the use of graphical network scanning tools.

9. Evidence Register

The following evidence should accompany this report:

1. PM5_Task1_Local_IP.png
2. PM5_Task2_Routing_Table.png
3. PM5_Task3_Gateway_Connectivity.png
4. PM5_Task4_Live_Host_Discovery.png
5. PM5_Task5_Zenmap_Profile.png
6. PM5_Task5_Zenmap_Results.png
7. PM5_Task6_Network_Topology.png
8. PM5_Task7_Topology_PDF_Evidence.png
9. Kali MAC address command screenshot.

Supporting files:

- PM5_Zenmap_Scan_Results.xml
- NETWORKWALKS_B082_W2_PM5_Network_Topology.pdf

End of Report